

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 7 Winter 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTMC701 - MECHATRONICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 The major drawback of the mechanical components in a mechatronic system is _____.
A) Fast operation speeds B) Strength
C) Wear and tear D) All of these
- 2 Which of the following principle can not be used in a proximity sensor?
A) Induction B) Optical C) Infrared D) Thermocouple
- 3 Capacitive sensors can be used for _____.
A) Force measurement B) Temperature measurement
C) Velocity D) Level
- 4 Which of the following system can be a typical example of a mechatronic system?
A) Robot B) ATM machine C) CNC D) All of these
- 5 A to D conversion refers to _____.
A) AC to DC current conversion B) Analogue to Digital conversion
C) Angle to degree conversion D) Automatic to Digital conversion
- 6 Typically in an Operational Amplifier, negative input is called as _____.
A) Inverting input B) Non-inverting input
C) Ground D) Bias
- 7 Zener Diode is used to protect the device from _____.
A) Wrong polarity B) High temperatures
C) High current D) Noise in the signal
- 8 Which of the following drive is preferred for extremely smooth motion?
A) Pneumatic B) Hydraulic C) Electrical D) None of these
- 9 PLC stands for _____.
A) Principle Large Controller B) Positive Linier Connector
C) Programmable Log Calculator D) None of these
- 10 Bode plot is used to understand the _____ of a system.
A) Efficiency B) Stability Response C) Capacity D) All of these

11 Which of the following is an example of On/Off type of controller?

- A) Water cooler B) Water geezer
- C) Air cooler D) Air Conditioner

12 The step size of a stepper motor depends upon _____.

- A) Voltage B) No of poles
- C) Current D) No of turns in stator windings

Q2. Solve the following.

- A) Explain the role of Mechanical and Electrical components in a Typical Mechtronic system with the help of suitable examples. 6
- B) Explain any three temperature sensors with their principle, advantages, limitations and applications. 6

Q3. Solve the following.

- A) Explain the inverting operational amplifier in details. 6
- B) What is the purpose of a protection device in an electronics circuit? What are the different types of protection devices used in typical mechtronic system. Explain Opto-isolator as a protection device. 6

Q4. Solve Any Two of the following.

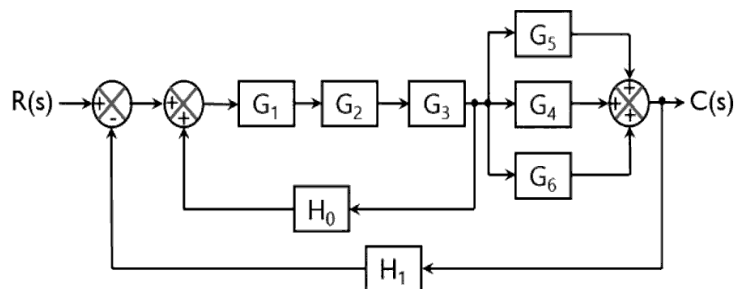
- A) Write short note on 'Servo motors'. 6
- B) Explain the classification of pneumatic valves based on number of ports and positions. 6
- C) Design a pneumatic circuit for achieving the following sequence of operation: $A + B + B - A -$ 6

Q5. Solve Any Two of the following.

- A) Differentiate between Microprocessor and Microcontroller. 6
- B) Explain the architecture of a typical microcontroller. 6
- C) Write a ladder logic for a level controller of a overhead tank. The pump should get started as soon as the level of water in the overhead tank falls below a certain level and it should stop at a certain level to avoid the overflow. 6

Q6. Solve Any Two of the following.

- A) Explain the open loop and closed loop control systems with suitable examples. 6
- B) What is a PID controller? Explain the advantages, limitations and applications of a PID controller. 6
- C) Find the overall transfer fuction for the following control system: 6



*** End ***